

Dependency-Aware Multi-Phase Liver Tumor Segmentation with Explicit Phase Interaction Modeling

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Abstract. Phase interaction in multi-phase CT is selective, not additive — yet existing segmentation models offer no mechanism to verify this claim or act on it during deployment. We propose *DependencyAttentionUNet*, a phase-aware architecture for joint liver and tumor segmentation from four-phase CT that preserves per-phase representations through a shared encoder, learns explicit per-phase importance gates, and models pairwise cross-phase interactions before decoding. On the complete Harvard multi-phase CT cohort (360/77/77 patient-wise split), the model achieves liver DSC 0.9492 and tumor DSC 0.8409. More importantly, it enables a complete combinatorial dependency analysis over all 15 non-empty phase subsets using a single trained model: the portal venous plus delayed (PVP+DP) two-phase subset achieves tumor DSC 0.8496 and the arterial plus PVP plus delayed (AP+PVP+DP) three-phase subset achieves 0.8533, both exceeding the all-phase result of 0.8409. This suggests that, under the fixed-model evaluation protocol used here, explicit phase-aware modeling provides interpretable phase-usage evidence together with flexible inference under reduced phase subsets without a large performance penalty — a finding inaccessible to any early-fusion baseline and directly relevant for clinical protocols with incomplete phase acquisition.

Keywords: Multi-Phase CT · Liver Tumor Segmentation · Phase Dependency · Phase Interaction Modeling · Interpretability

1 Introduction

Liver cancer is among the most prevalent and lethal malignancies worldwide [18], and accurate liver and tumor segmentation from CT is a prerequisite for treatment planning, surgical guidance, and longitudinal monitoring [5]. In clinical practice, multi-phase CT acquires the liver under non-contrast (NC), arterial (AP), portal venous (PVP), and delayed (DP) conditions, each providing distinct enhancement characteristics and lesion conspicuity. The relative signal between phases carries diagnostic information central to LI-RADS lesion characterisation [5] that single-phase imaging cannot replicate.

Deep learning segmentation models based on U-Net [16] and its variants have achieved strong results on single-phase benchmarks such as LiTS [2]. Extending these models to multi-phase CT typically involves concatenating phase channels or applying attention-based fusion [20,12,22]. Such approaches exploit multi-phase evidence but do not treat phase relationships as a primary object of learning: the model cannot communicate whether a given phase is informative, redundant, or complementary relative to others [22]. Recent work on domain-adaptive four-phase transformers [14] and phase-attention fusion networks [12] has moved toward more explicit phase treatment, but neither provides a systematic analysis of which phase subsets are informative.

We address this gap by proposing a *DependencyAttentionUNet* that (i) preserves a distinct latent representation for each phase through a shared per-phase encoder, (ii) learns explicit per-phase importance gates, and (iii) models pairwise cross-phase interactions before the joint decoder. We evaluate the model against three controlled trained baselines (RIU-Net [13], Mixed U-Net, and SPA-UNet [9]) on the same patient-wise held-out test set, and perform a complete combinatorial dependency analysis over all 15 non-empty phase subsets.

Contributions.

1. A phase-aware segmentation architecture with shared per-phase encoding, learned importance gates, and explicit pairwise interaction modeling, yielding interpretable phase-usage evidence as a training by-product.
2. A comprehensive same-dataset baseline comparison spanning three controlled multi-phase segmentation baselines trained under identical conditions.
3. A complete combinatorial dependency analysis over all 15 non-empty phase subsets, showing under a fixed-model evaluation protocol that phase interaction is selective and that certain reduced subsets can outperform the all-phase configuration.

2 Related Work

Liver and tumor segmentation. U-Net [16] established the encoder–decoder paradigm for dense medical segmentation, and subsequent work explored denser skip connections, residual blocks, and attention mechanisms [10,15,7]. Lightweight 2.5D variants such as RIU-Net [13] and UV-Net [21] capture axial context without full 3D cost. Transformer-based models including TransUNet [4] and Swin-Unet [3] extend the paradigm to global self-attention and have shown strong abdominal segmentation performance. More recent attention U-Net variants including UCTransNet [19], SPA-UNet [9], and RHEU-Net [17] further refine tumor delineation through channel-wise transformer skip connections, spatial pyramid pooling, and hybrid gated attention respectively, all within a single-stream framework. These methods form relevant comparison points for the present work because they remain single-stream segmentation architectures and therefore do not expose explicit phase-dependency modeling.

Multi-phase CT fusion. Methods for multi-phase CT liver analysis have explored channel fusion, phase attention, and spatial aggregation [20,12,22]. Prior phase-aware segmentation methods treat phase selection implicitly: PA-ResSeg [20] and PA-Net [12] learn phase attention weights but do not expose which phases drive the prediction, while DA-Tran [14] addresses cross-phase domain shift via transformer fusion. Complementary directions include evidential uncertainty for ambiguous phase labels [6], adaptive multi-scale supervision across phases [8]. A thread common to these segmentation-focused approaches is that phase utility is treated as a training-time property and remains opaque at inference; none provides a systematic post-hoc analysis of which phase subsets are informative for a trained model.

3 Proposed Method

3.1 Dataset and Preprocessing

We use an institutional Harvard multi-phase liver CT cohort comprising 514 patients with four co-registered CT phases (NC, AP, PVP, DP) and corresponding liver and tumor annotation masks. This cohort is distinct from the public LiTS benchmark [2]; it is not publicly available and was accessed under a data-use agreement. The cohort is split patient-wise into 360 training, 77 validation, and 77 test subjects, with no patient overlap across splits.

Each training example uses a 2.5D representation: three consecutive axial slices per phase yield 12 input channels (4×3), preserving local axial context without requiring full 3D computation. All volumes are clipped to the Hounsfield Unit range $[-200, 250]$ and normalised to $[0, 1]$. Phase identification, cache management, and preprocessing follow a reproducible NPZ-based pipeline. Tumor-aware oversampling with weight 12.0 ensures sufficient exposure to tumor-containing slices during training; the 360-patient training set yields approximately 9,200 tumor-positive and 13,100 tumor-negative slices after liver-slice filtering.

Reproducibility details. All models use the same cached NPZ pipeline, patient-wise split, and 2.5D input construction. Each slice is resized to 256×256 , and each input is formed from three consecutive axial slices per phase (centre slice ± 1 neighbour), giving 12 channels in total across NC, AP, PVP, and DP. Training, validation, and test partitions are fixed once at the patient level with no patient overlap across splits. CT intensities are clipped to $[-200, 250]$ before min-max normalisation. Reported liver DSC and tumor DSC are computed after final thresholding and post-processing on held-out test slices, then averaged over the test set; all baseline and subset comparisons in this paper follow this same evaluation definition.

3.2 DependencyAttentionUNet Architecture

The proposed architecture processes the four phases through a shared per-phase encoder before fusing their representations, departing from multi-channel base-

lines that concatenate all phases into a single input stream and thereby lose phase identity after the first convolutional layer.

Shared per-phase encoder. The 12-channel input is split into four groups of three (one per phase) and each group is passed through a common encoder with four convolutional stages and a bottleneck. Weight sharing across phases encourages a common representation space while preserving the identity of each phase’s feature map and skip tensors, enabling direct comparison of phase representations after encoding.

Phase importance gates. A scalar gate $g_k \in [0.15, 1.0]$ is computed per phase from its bottleneck feature via global average pooling and a sigmoid projection. The lower clamp prevents any phase from collapsing to zero contribution. The decoder receives a gate-weighted average of the four bottleneck features and skip tensors:

$$\begin{aligned} \mathbf{f}_{\text{fused}} &= \frac{\sum_k g_k \mathbf{f}_k}{\sum_k g_k}, \\ \mathbf{s}_{\text{fused}}^{(l)} &= \frac{\sum_k g_k \mathbf{s}_k^{(l)}}{\sum_k g_k}, \end{aligned} \quad (1)$$

where $\mathbf{f}_k$ and $\mathbf{s}_k^{(l)}$ are the bottleneck and l -th skip features of phase k .

Explicit pairwise interaction. For each pair (i, j) , an interaction feature $\mathbf{h}_{ij} = \mathbf{f}_i \odot \mathbf{f}_j$ is computed via elementwise multiplication. The $\binom{4}{2} = 6$ interaction maps are averaged and projected back into the bottleneck space via a 1×1 convolution before being added to $\mathbf{f}_{\text{fused}}$. This explicitly encodes where phases agree or jointly highlight lesion-relevant structures.

Decoder. The decoder mirrors the encoder depth and uses attention-gated skip connections [15], selectively suppressing non-salient features before concatenation.

3.3 Training Objective and Inference

The training loss combines focal cross-entropy [11] ($\lambda_1 = 0.25$), class-weighted Dice loss ($\lambda_2 = 0.35$), and tumor focal Tversky loss [1] ($\lambda_3 = 0.40$). The Tversky term up-weights false negatives, the dominant failure mode for small tumors. Checkpoints are selected by $0.35 \cdot \text{DSC}_{\text{liver}} + 0.65 \cdot \text{DSC}_{\text{tumor}}$, prioritising tumor without abandoning liver quality.

All models are trained with AdamW (LR 10^{-4} , weight decay 10^{-3}) for up to 24 epochs with early stopping (patience 5). During inference we apply horizontal-flip TTA, validation-tuned per-class thresholds (liver 0.50, tumor 0.20), minimum connected-component filtering (32 tumor pixels), and liver-constrained tumor decoding.

Table 1. Segmentation performance on the Harvard test set (77 patients). All trained models share the same 360/77/77 split and inference policy.

Method	Family	Liver DSC	Tumor DSC
RIU-Net [13]	2.5D Res-Inception U-Net	0.9298	0.8137
Mixed U-Net	Hybrid 2D/3D U-Net	0.9431	0.8570
SPA-UNet [9]	Spatial pyramid U-Net	0.9594	0.8458
Proposed	DependencyAttentionUNet	0.9492	0.8409

4 Experiments

4.1 Baselines

All trained baselines share the identical 360/77/77 patient-wise split, preprocessing pipeline, 2.5D input format, loss function, optimizer, checkpoint-selection rule, validation-tuned thresholds, test-time augmentation, post-processing, and liver-constrained tumor decoding. Architecture is the sole variable.

- **RIU-Net** [13]: Lightweight 2.5D residual-Inception U-Net. Strong parameter-efficient 2.5D baseline.
- **Mixed U-Net**: Controlled Harvard-side reproduction and adaptation of a hybrid 2D/3D U-Net design, implemented to operate under the same 12-channel 2.5D input and shared evaluation pipeline used throughout this study.
- **SPA-UNet** [9]: Spatial pyramid pooling and fused multi-scale attention U-Net for liver tumor delineation.

5 Results

5.1 Comparison with Baselines

Table 1 presents segmentation performance on the 77-patient held-out test set. For broader context rather than as a controlled baseline, our earlier single-phase U-Net adaptation on the public LiTS benchmark [2] achieved liver DSC 0.9111 and tumor DSC 0.3360, illustrating the gap between single-phase and multi-phase settings in our project family. Among the Harvard-trained models, the proposed *DependencyAttentionUNet* achieves tumor DSC 0.8409 and liver DSC 0.9492. RIU-Net reaches tumor DSC 0.8137 (−2.7 pp), confirming that explicit dependency modeling improves over a directly comparable 2.5D baseline. Mixed U-Net achieves tumor DSC 0.8570, surpassing the proposed model’s all-phase result. However, Mixed U-Net provides no mechanism for phase attribution or subset analysis; the central contribution of the proposed model lies in interpretable phase-usage evidence and flexible phase-subset inference rather than raw all-phase superiority alone (see Sect. 5.4).

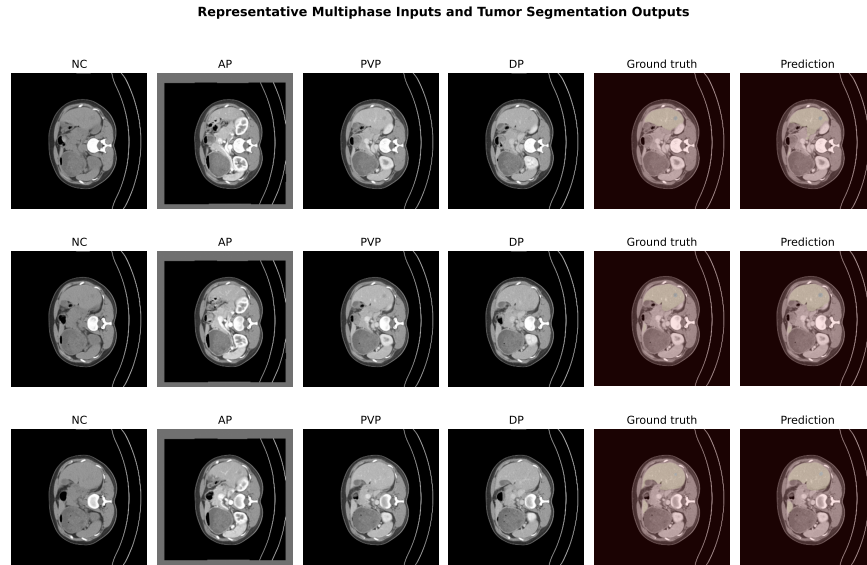


Fig. 1. Representative segmentation examples from the held-out test split. Each row shows multi-phase input slices, the ground-truth three-class mask (background / liver / tumor), and the proposed model prediction after test-time augmentation, threshold tuning, and post-processing.

5.2 Training Dynamics

Figure 3 shows the training dynamics of the proposed model on the full study cohort. Validation tumor DSC stabilises and triggers early stopping at epoch 23 (best checkpoint: epoch 18), while liver DSC plateaus steadily above 0.94.

5.3 Phase Importance Gates

Figure 4 shows the average per-phase gate activations on the test set: NC = 0.274, AP = 0.259, PVP = 0.250, DP = 0.259. The near-uniform distribution suggests that, in the 2.5D setting used here, all four phases retain useful information after local axial stacking. At the same time, the non-trivial variation indicates that the model does not reduce to a simple unweighted average: NC receives the highest average gate value (0.274), while PVP receives the lowest (0.250) despite being the strongest single phase in Table 2. One plausible interpretation is that PVP contributes strong complementary evidence when evaluated alone or in small subsets, whereas NC may act as a stable anatomical reference in the all-phase setting. The inter-patient standard deviation of each gate is below 0.03, supporting the view that the learned phase preferences are consistent at the test-set level. These activations emerge from the segmentation objective alone with no direct phase-importance supervision.

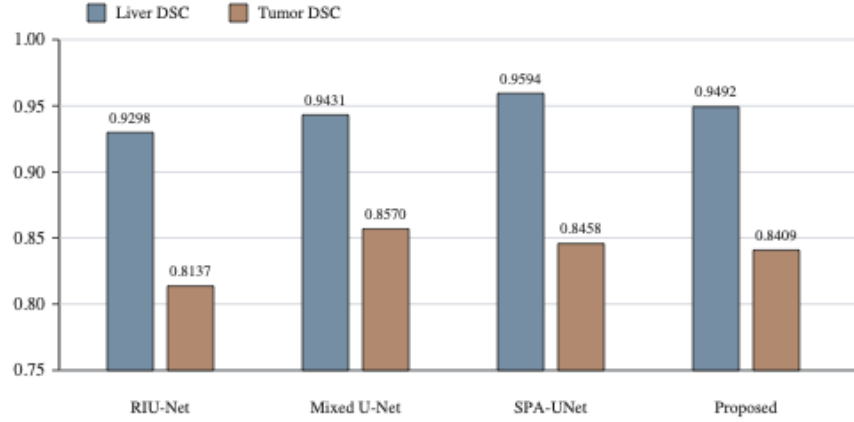


Fig. 2. Liver and Tumor DSC across all baselines on the 77-patient Harvard test set. All trained models share the same 360/77/77 patient-wise split and inference policy.

5.4 Phase Dependency Analysis

Table 2 presents the complete combinatorial analysis over all 15 non-empty phase subsets, evaluated on the *fixed* trained model — that is, a single model trained on all four phases, with phases masked at inference time only. To make the subset ranking easier to interpret visually, Fig. 5 orders all 15 subset tumor DSC scores from strongest to weakest, while Fig. 6 summarises the best subset at each cardinality. Three findings stand out.

Subset inference protocol. For a chosen subset, channels belonging to unavailable phases are zeroed before the forward pass and the corresponding phase-presence indicators are set to zero as well. The masked phases therefore contribute neither bottleneck features nor skip features nor gate weight to the fused representation. Fusion is recomputed only over the retained phases, and the same trained decoder, thresholds, test-time augmentation, and post-processing pipeline are kept fixed. In this way, all 15 non-empty phase subsets are evaluated under a single consistent inference procedure without retraining separate models.

PVP dominates single-phase performance. PVP is the strongest single phase (tumor DSC 0.5360), more than 13 percentage points ahead of AP (0.3960), consistent with its clinical role in portal venous contrast differentiation [5].

Two-phase combinations exhibit strong non-linearity. The best two-phase subset PVP+DP achieves tumor DSC 0.8496, a gain of +31 pp over PVP alone. This non-linear jump is observable only because the model preserves phase identity and models explicit pairwise interactions; an early-fusion model cannot isolate this behaviour.

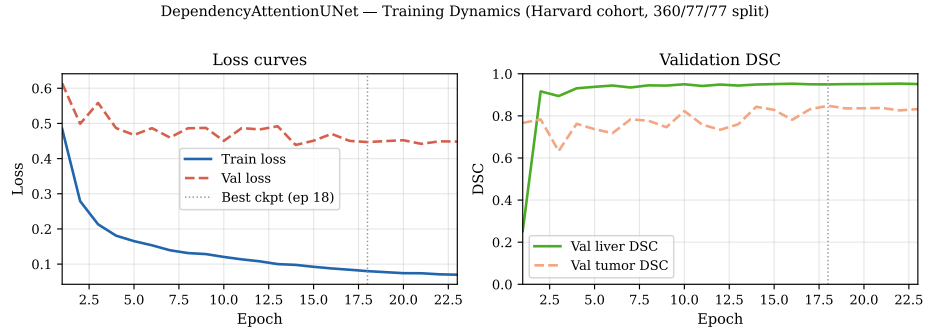


Fig. 3. Training dynamics of the *DependencyAttentionUNet* on the full study cohort. Best checkpoint (epoch 18) is selected by $0.35 \cdot \text{DSC}_{\text{liver}} + 0.65 \cdot \text{DSC}_{\text{tumor}}$. Early stopping triggers at epoch 23.

Reduced subsets can outperform the all-phase configuration. Both PVP+DP (0.8496) and AP+PVP+DP (0.8533) exceed the all-phase result (0.8409). This indicates that, for this trained model, reduced-phase inference can remain competitive and may even score higher for certain subsets — a notable result given that NC is the phase with the highest average gate activation. A practical consequence of the gated fusion design is that the model can run on any non-empty phase subset at inference without retraining; unavailable phases are excluded from the gate-weighted average and the remaining gates redistribute the weighting automatically. This is directly relevant for clinical settings where not all phases are acquired, as the model degrades gracefully rather than failing.

6 Discussion

Phase selectivity as the primary finding. The most significant result of this work is not the absolute DSC score but the empirical demonstration that phase interaction is selective: PVP+DP (0.8496) and AP+PVP+DP (0.8533) both exceed the all-phase result (0.8409) under this fixed-model evaluation protocol. Adding NC to either of the two best three-phase subsets does not improve tumor DSC in these experiments. This finding is only accessible because the model preserves phase identity and allows arbitrary phase masking at inference time — properties that no early-fusion baseline possesses. For a model trained on four-phase data, AP+PVP+DP scores slightly higher than the full four-phase configuration, a finding relevant to clinical settings where reducing acquisition phases would lower patient dose and scan time. Prospective validation with models trained specifically on reduced-phase data would be needed to generalise this observation to a protocol recommendation.

Comparison with concatenation baselines. Mixed U-Net achieves the highest raw tumor DSC (0.8570) among trained models, confirming that well-tuned multi-channel concatenation is competitive for all-phase inference. However, Mixed

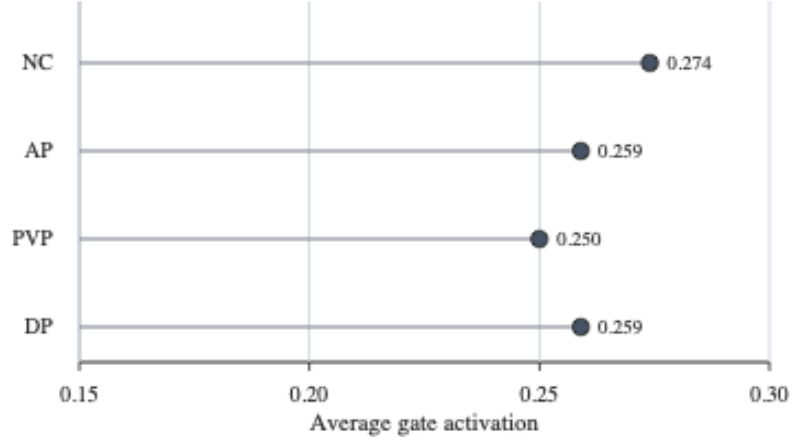


Fig. 4. Average phase-gate activations on the 77-patient test set. Gates are clamped to $[0.15, 1.0]$ during training. The near-uniform distribution is consistent with broadly shared phase utility in this 2.5D setting, with a slight test-set preference toward NC.

U-Net provides no mechanism to identify which phases matter, to operate under missing phases, or to explain phase contributions. The proposed model trades a small raw-Dice gap for a fundamentally different and richer capability set: per-phase gate weights, arbitrary phase-subset inference, and the complete 15-subset dependency analysis. These are architectural properties that no single-stream baseline can replicate regardless of its Dice score.

Limitations. (i) All trained baselines share one Harvard patient-wise split; no external validation cohort is included. (ii) Evaluation is slice-level; volumetric DSC aggregated per patient may yield different rankings and will be computed in follow-up work. (iii) The 15-subset analysis is performed on a *fixed* trained model; retraining on each subset could alter subset rankings and is left for future study. (iv) The pairwise interaction term does not explicitly model three- or four-way phase coupling, which may limit expressiveness for subsets where higher-order interactions dominate. (v) A component-wise ablation study isolating the individual contributions of the shared encoder, phase gates, and pairwise interaction term is left for future work. (vi) Results are reported for one patient-wise split and one training run per model; confidence intervals, repeated-seed experiments, and formal significance testing would strengthen the robustness of the observed subset rankings.

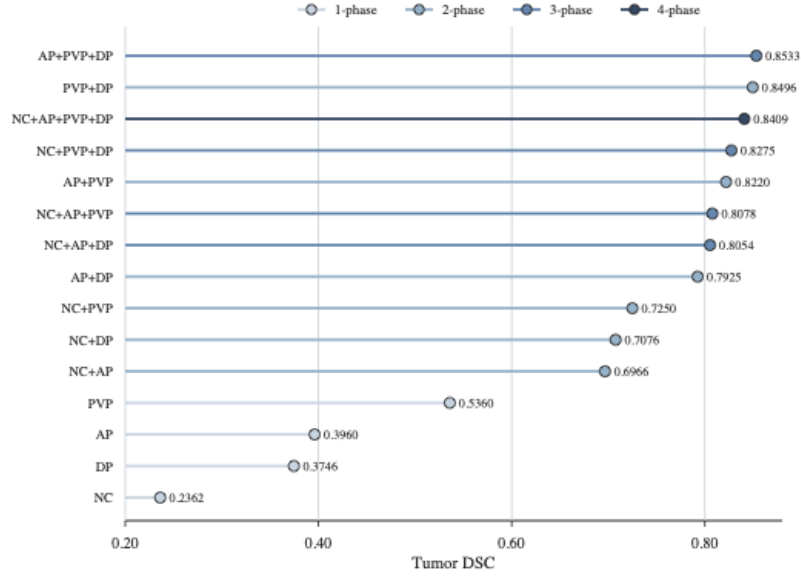


Fig. 5. Tumor DSC for all 15 non-empty phase subsets, ranked from highest to lowest, under the fixed trained-model evaluation protocol. Color indicates subset cardinality. The plot makes the selective benefit of particular two- and three-phase combinations visually explicit.

7 Conclusion

We demonstrated that phase interaction in multi-phase CT liver tumor segmentation is selective rather than uniformly additive — a finding verifiable only through explicit phase-aware modeling. The proposed *DependencyAttentionUNet* enables this through a shared per-phase encoder, learned importance gates, and explicit pairwise cross-phase interaction, and yields liver DSC 0.9492 and tumor DSC 0.8409 on the full Harvard multi-phase CT study cohort. The complete combinatorial analysis over all 15 non-empty phase subsets, conducted on the fixed trained model with phases masked at inference, reveals that PVP+DP achieves tumor DSC 0.8496 and AP+PVP+DP achieves 0.8533, both exceeding the all-phase result of 0.8409. This indicates that, for this trained model, the main value of explicit phase-aware modeling is not raw all-phase superiority alone, but interpretable phase usage together with flexible inference under reduced phase subsets — a result directly relevant to clinical deployment under variable acquisition protocols. Future work will extend to three- and four-way interaction terms, volumetric evaluation metrics, and prospective validation on protocols with naturally incomplete phase acquisition.

Table 2. Complete phase-subset dependency analysis on the fixed trained model. All 15 non-empty subsets evaluated on the 77-patient test set. Bold marks the top tumor DSC within each cardinality group.

Input phases	N	Liver DSC	Tumor DSC
NC	1	0.8476	0.2362
AP	1	0.9099	0.3960
PVP	1	0.9500	0.5360
DP	1	0.9004	0.3746
NC+AP	2	0.9228	0.6966
NC+PVP	2	0.9364	0.7250
NC+DP	2	0.9097	0.7076
AP+PVP	2	0.9546	0.8220
AP+DP	2	0.9358	0.7925
PVP+DP	2	0.9507	0.8496
NC+AP+PVP	3	0.9491	0.8078
NC+AP+DP	3	0.9342	0.8054
NC+PVP+DP	3	0.9447	0.8275
AP+PVP+DP	3	0.9522	0.8533
NC+AP+PVP+DP	4	0.9492	0.8409

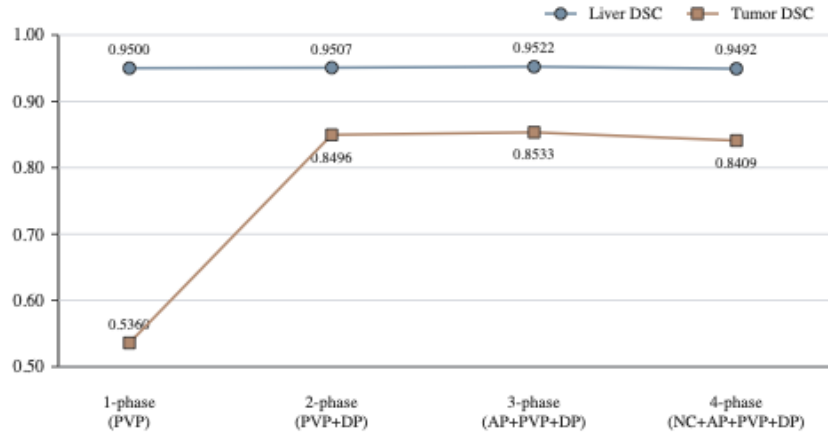


Fig. 6. Best-performing phase subset within each subset cardinality. Liver DSC and tumor DSC are shown together to highlight how tumor performance improves sharply from one to two phases, peaks at the three-phase AP+PVP+DP subset, and slightly declines for the full four-phase configuration.

Acknowledgments. The authors acknowledge the contributors of the Harvard multi-phase liver CT dataset. Code can be accessed at <https://github.com/areebanaveed-12/Phase-Dependency-Liver-Segmentation>.

Disclosure of Interests. The authors have no competing interests to declare.

References

1. Abraham, N., Khan, N.M.: A novel focal Tversky loss function with improved attention U-Net for lesion segmentation. In: IEEE International Symposium on Biomedical Imaging (ISBI). pp. 683–687 (2019). <https://doi.org/10.1109/ISBI.2019.8759329>
2. Bilic, P., Christ, P., Li, H.B., Vorontsov, E., Ben-Cohen, A., Kaissis, G., Szeskin, A., Jacobs, C., Mamani, G.E.H., Chartrand, G., et al.: The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis* **84**, 102680 (2023). <https://doi.org/10.1016/j.media.2022.102680>
3. Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., Wang, M.: Swin-Unet: Unet-like pure transformer for medical image segmentation. In: Computer Vision – ECCV 2022 Workshops. Lecture Notes in Computer Science, vol. 13803, pp. 205–218. Springer (2022). https://doi.org/10.1007/978-3-031-25066-8_9
4. Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y.: TransUNet: Transformers make strong encoders for medical image segmentation. arXiv preprint arXiv:2102.04306 (2021), arXiv:2102.04306
5. Chernyak, V., Fowler, K.J., Kamaya, A., Kielar, A.Z., Elsayes, K.M., Bashir, M.R., Kono, Y., Do, R.K., Mitchell, D.G., Singal, A.G., Tang, A., Sirlin, C.B.: Liver imaging reporting and data system (LI-RADS) version 2018: Imaging of hepatocellular carcinoma in at-risk patients. *Radiology* **289**(3), 816–830 (2018). <https://doi.org/10.1148/radiol.2018181494>
6. Hu, C., Xia, T., Cui, Y., Zou, Q., Wang, Y., Xiao, W., Ju, S., Li, X.: Trustworthy multi-phase liver tumor segmentation via evidence-based uncertainty. *Engineering Applications of Artificial Intelligence* **133**, 108289 (2024). <https://doi.org/10.1016/j.engappai.2024.108289>
7. Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H.: nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* **18**(2), 203–211 (2021). <https://doi.org/10.1038/s41592-020-01008-z>
8. Kuang, H., Yang, X., Li, H., Wei, J., Zhang, L.: Adaptive multi-phase liver tumor segmentation with multi-scale supervision. *IEEE Signal Processing Letters* **31**, 426–430 (2024). <https://doi.org/10.1109/LSP.2024.3356414>
9. Li, W., Jia, M., Yang, C., Lin, Z., Yu, Y., Zhang, W.: SPA-UNet: A liver tumor segmentation network based on fused multi-scale features. *Open Life Sciences* **18**(1), 20220685 (2023). <https://doi.org/10.1515/biol-2022-0685>
10. Li, X., Chen, H., Qi, X., Dou, Q., Fu, C.W., Heng, P.A.: H-DenseUNet: Hybrid densely connected UNet for liver and tumor segmentation from CT volumes. *IEEE Transactions on Medical Imaging* **37**(12), 2663–2674 (2018). <https://doi.org/10.1109/TMI.2018.2845918>
11. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: IEEE International Conference on Computer Vision (ICCV). pp. 2980–2988 (2017). <https://doi.org/10.1109/ICCV.2017.324>

12. Liu, Z., Hou, J., Pan, X., Zhang, R., Shi, Z.: PA-Net: A phase attention network fusing venous and arterial phase features of CT images for liver tumor segmentation. *Computer Methods and Programs in Biomedicine* **244**, 107997 (2024). <https://doi.org/10.1016/j.cmpb.2023.107997>
13. Lv, P., Wang, J., Wang, H.: 2.5D lightweight RIU-Net for automatic liver and tumor segmentation from CT. *Biomedical Signal Processing and Control* **75**, 103567 (2022). <https://doi.org/10.1016/j.bspc.2022.103567>
14. Ni, Y., Chen, G., Feng, Z., Cui, H., Metaxas, D., Zhang, S., Zhu, W.: DA-Tran: Multiphase liver tumor segmentation with a domain-adaptive transformer network. *Pattern Recognition* **149**, 110233 (2024). <https://doi.org/10.1016/j.patcog.2023.110233>
15. Oktay, O., Schlemper, J., Le Folgoc, L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N.Y., Kainz, B., Glocker, B., Rueckert, D.: Attention U-Net: Learning where to look for the pancreas. In: *Medical Imaging with Deep Learning (MIDL)* (2018), <https://openreview.net/forum?id=Skft7cijm>, mIDL 2018; available as OpenReview preprint
16. Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. pp. 234–241. Springer (2015). https://doi.org/10.1007/978-3-319-24574-4_28
17. Sun, L., Jiang, L., Wang, M., Wang, Z., Xin, Y.: A multi-scale liver tumor segmentation method based on residual and hybrid attention enhanced network with contextual integration. *Sensors* **24**(17), 5845 (2024). <https://doi.org/10.3390/s24175845>
18. Sung, H., Ferlay, J., Siegel, R.L., Laversanne, M., Soerjomataram, I., Jemal, A., Bray, F.: Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians* **71**(3), 209–249 (2021). <https://doi.org/10.3322/caac.21660>
19. Wang, H., Cao, P., Wang, J., Zaiane, O.R.: UCTransNet: Rethinking the skip connections in U-Net from a channel-wise perspective with transformer. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 36, pp. 2441–2449 (2022). <https://doi.org/10.1609/aaai.v36i3.20144>
20. Xu, Y., Cai, M., Lin, L., Zhang, Y., Hu, H., Peng, Z., Zhang, Q., Chen, Q., Mao, X., Iwamoto, Y., Han, X.H., Chen, Y.W., Tong, R.: PA-ResSeg: A phase attention residual network for liver tumor segmentation from multiphase CT images. *Medical Physics* **48**(7), 3752–3766 (2021). <https://doi.org/10.1002/mp.14922>
21. Zhang, C., Hua, Q.Q., Chu, Y.Y., Wang, P.W.: Liver tumor segmentation using 2.5D UV-Net with multi-scale convolution. *Computers in Biology and Medicine* **133**, 104424 (2021). <https://doi.org/10.1016/j.combiomed.2021.104424>
22. Zhang, Y., Peng, C., Peng, L., Huang, H., Tong, R., Lin, L., Li, J., Chen, Y.W., Chen, Q., Hu, H., Peng, Z.: Multi-phase liver tumor segmentation with spatial aggregation and uncertain region inpainting. In: *Medical Image Computing and Computer Assisted Intervention (MICCAI)*. pp. 68–77. Springer (2021). https://doi.org/10.1007/978-3-030-87193-2_7